

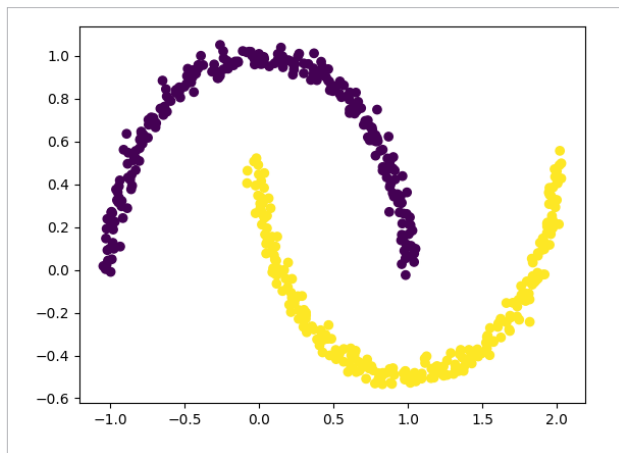


Figure 7: Distribution of moons data set

Figure 7 shows the distribution of the original moons data set. It can be seen that the data is non-linearly separable. Now let us try to apply PCA on this data:

```
moon_pca = PCA(n_components = 2)
X_pca = moon_pca.fit_transform(train_X)

plt.title("Moon PCA")
plt.scatter(X_pca[:, 0], X_pca[:, 1], c = train_y*10)
#plt.scatter(X_pca[:, 0], X_pca[:, 1], color="#88c999")
plt.xlabel("PCA Component 1")
plt.ylabel("PCA Component 2")
plt.show()
```

Figure 8 shows the distribution of data after applying PCA. It can be seen that data is still not separable and needs transformation. Let us now apply the Kernel PCA method from *sklearn* and check the results:

```
moon_kpca = KernelPCA(kernel = 'rbf', gamma = 15)
X_kpca = moon_kpca.fit_transform(train_X)

plt.title("Kernel PCA")
plt.scatter(X_kpca[:, 0], X_kpca[:, 1], c = train_y*10)
plt.show()
```

From Figure 9 it is clearly visible that Kernel PCA is able to successfully make the data separable.

To sum up, dimensionality reduction techniques are used to solve the problem of curse of dimensionality. Each method has its own advantages, and can be used based on the characteristics of the data samples, like being linearly separable or being non-linearly separable. Most of the dimensionality reduction techniques are readily available in the *sklearn* library package, and can be applied and visualised. By reducing the dimensions of original data, these

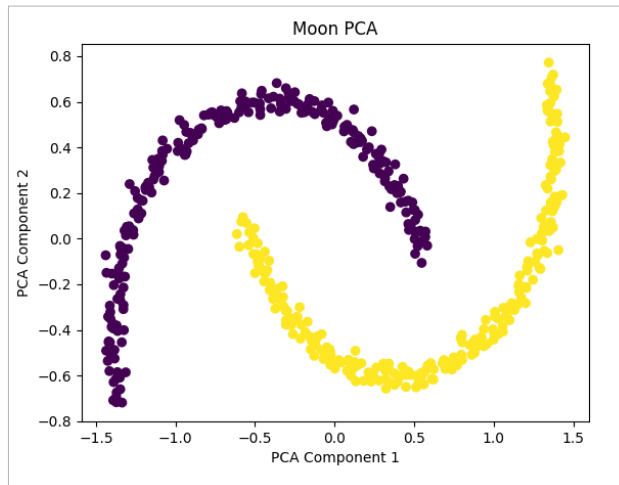


Figure 8: PCA for moons data set

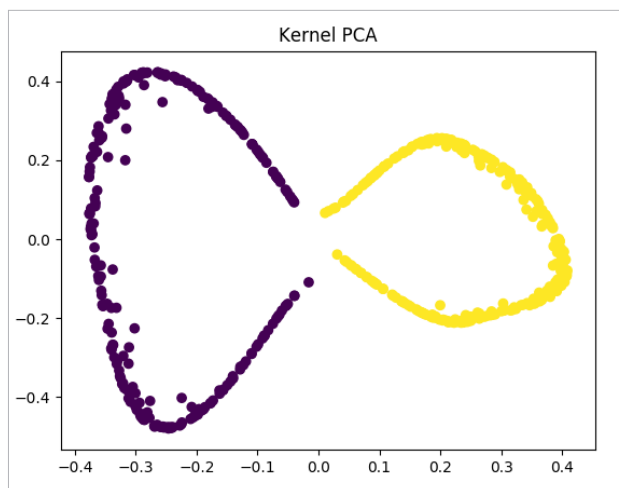


Figure 9: Kernel PCA for moons data set

techniques reduce computing costs, improve performance and enable insights into data. **END** 🐧

Reference

<https://scikit-learn.org/>

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